PROJECT REPORT

SOLAR LOAD FILTERS

ABSTRACT:

If we take into consideration the current situation from the world's point of view then we can say that the world today is greatly dependent on fossil fuels and other conventional sources of energy. The exaggerated rate of consumption of fossil fuels has produced adverse effects on environment and their increasing cost have hampered the economic situation of every nation all over the world. India consumed 136 million tonnes of oil in 2007 of which only 34 million tonnes have been produced in India. This alone shows the need for non-conventional energy sources.

Over the last 50 years, India is continuously replacing non-commercial energy sources by commercial energy sources such as coal, oil, gas and nuclear. The energy share of conventional energy sources is expected to go up to 76.5% by 2011-2012 from 29% in 1953-1954. Share of oil in total consumption has also grown significantly from mere 5% in 1960 to 25% in 2007 of the global average. Presently India is largely dependent upon import of crude oil to the extent of 72% of the total demand.

India is one of the fastest growing consumers of conventional energy sources. Considering that India spends 24 billion dollars on the imports of crude oil, it is necessary to conserve non-renewable energy that is available to us. There is a need of complete independence from conventional energy sources. That is what our project focuses on.

Our project focuses on complete independence of an entire complex from conventional energy sources.

INTRODUCTION:

1.Project Objectives:

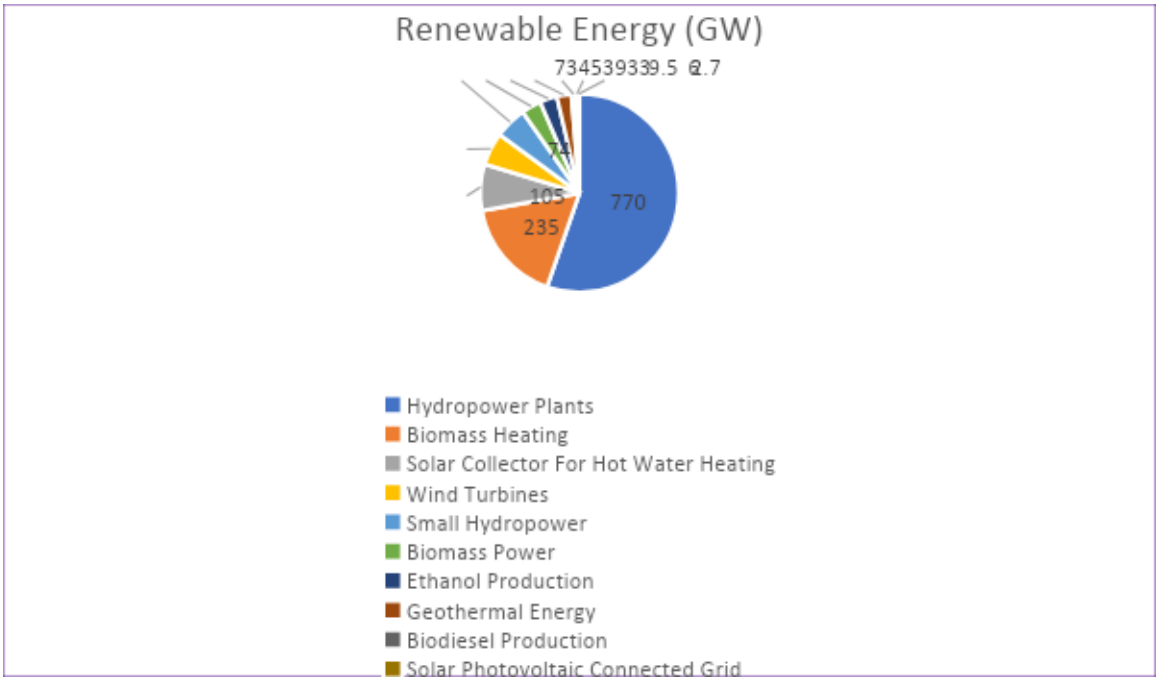
The plan is to design such a complex which completely runs on non-conventional sources of energy.

As the number of power cuts in India have considerably increased, we need to find an alternative solution for energy supply. Non-conventional source of energy can serve as a good alternative to conventional method. But there are a number of difficulties such as transmission problem. In transmission 60% of energy is lost. Renewable energy can be particularly suitable for developing countries like India. In rural and remote areas, transmission and distribution of energy generated from fossil fuels can be difficult and expensive. Producing renewable energy locally can offer a viable alternative.

Renewable energy projects in many developing countries have demonstrated that renewable energy can directly contribute to poverty alleviation by providing the energy needed for creating businesses and employment. Renewable energy technologies can also make indirect contributions to alleviating poverty by providing energy for cooking, space heating, and lighting. Renewable energy can also contribute to education, by providing electricity to schools. So, we decided to run an entire complex on conventional energy source. The solar panels which are already used in lighting homes can be improved by running a number of

applications on it. The biomass energy, which is a waste in today's era is also used for cooking in some areas. But both the energy resources are never used in combination. This type of arrangement if made would result in the conservation of energy along with its management. This system is best suited for the cafes, malls on the highways or in remote areas where it is difficult for energy transport.

The following diagram shows energy production at the end of year 2009:



This enormous energy of the sun cannot be used completely. What are the limitations and the methods associated to overcome it are discussed further in this project.

The energy management is not proper in India. India is both densely populated and has high solar insolation, providing an ideal combination for solar power in India. Much of the country does not have an electric grid, so one of the first applications of solar power has been for water pumping, to begin replacing India's four to five million diesel powered water pumps, each consuming about 3.5 kilowatts, and off-grid lighting. Some large projects have been proposed, and a 35,000 km^2 area of the That Desert has been set aside for solar power projects, sufficient to generate 700 to 2,100 gigawatts.

Lack of electricity infrastructure is one of the main hurdles in the development of rural India. India's grid system is considerably under-developed, with major sections of its populace still surviving off-grid. As of 2004 there are about 80,000 unelectrified villages in the country. Of these villages, 18,000 could not be electrified through extension of the conventional grid. A target for electrifying 5,000 such villages was fixed for the Tenth National Five-Year Plan 2002-2007). As of 2004, more than 2,700 villages had been electrified mainly comprising of SPV systems. Developments on cheap solar technology is considered as a potential alternative that allows an electricity infrastructure comprising of a network of local-grid clusters with distributed electricity generation That could allow bypassing, or at least relieving the need of installing expensive and long-distance centralized power delivery systems and yet bring cheap electricity to the masses.

Thus, application of solar is possible in India. There is just need of alarming the people of energy conservation and management.

2.Introduction to Solar Terminology:

Solar energy is the light and radiant heat from the Sun that influences Earth's climate and weather and sustains life.

Solar power is the rate of solar energy at a point in time; it is sometimes used as a synonym for solar energy or more specifically to refer to electricity generated from solar radiation.

Since ancient times solar energy has been harnessed for human use through a range of technologies.

Solar radiation along with secondary solar resources such as wind and wave power, hydroelectricity and biomass account for most of the available flow of renewable energy on Earth.

Solar energy technologies can provide electrical generation by heat engine or photovoltaic means, day lighting and space heating in passive solar and active solar buildings, potable water via distillation and disinfection, hot water, space cooling by absorption or vapor-compression refrigeration, thermal energy for cooking, and high temperature process heat for industrial purposes.

3.Solar Radiation:

Sunlight, in the broad sense, is the total spectrum of the electromagnetic radiation given off by the Sun.

On Earth, sunlight is filtered through the atmosphere, and the solar radiation is obvious as daylight when the Sun is above the horizon.

This is usually during the hours known as day. Near the poles in summer, sunlight also occurs during the hours known as night and in the winter at the pole’s, sunlight may not occur at any time.

When the direct radiation is not blocked by the clouds, it is experienced in the sunshine, a combination of bright light and heat.

Radiant heat directly produced by the radiation of the sun differs because of the increase in atmospheric temperature due to the radiative heating of the atmosphere by the sun's radiation.

Sunlight may be recorded using either pyranometer or pyrheliometer. Direct sunlight gives about 93 lumens of illumination per watt of electromagnetic power, including infrared, visible, and ultra-violet.

The Earth receives 174 petawatts (PW) of incoming solar radiation (insolation) at the upper atmosphere. Approximately 30% is reflected back to space while the rest is absorbed by the clouds, oceans and land masses.

The spectrum of solar light at the Earth's surface is mostly spread across the visible region.

A small part within the ultraviolet region is absorbed and this absorbed solar light heats the land surface, oceans and atmosphere.

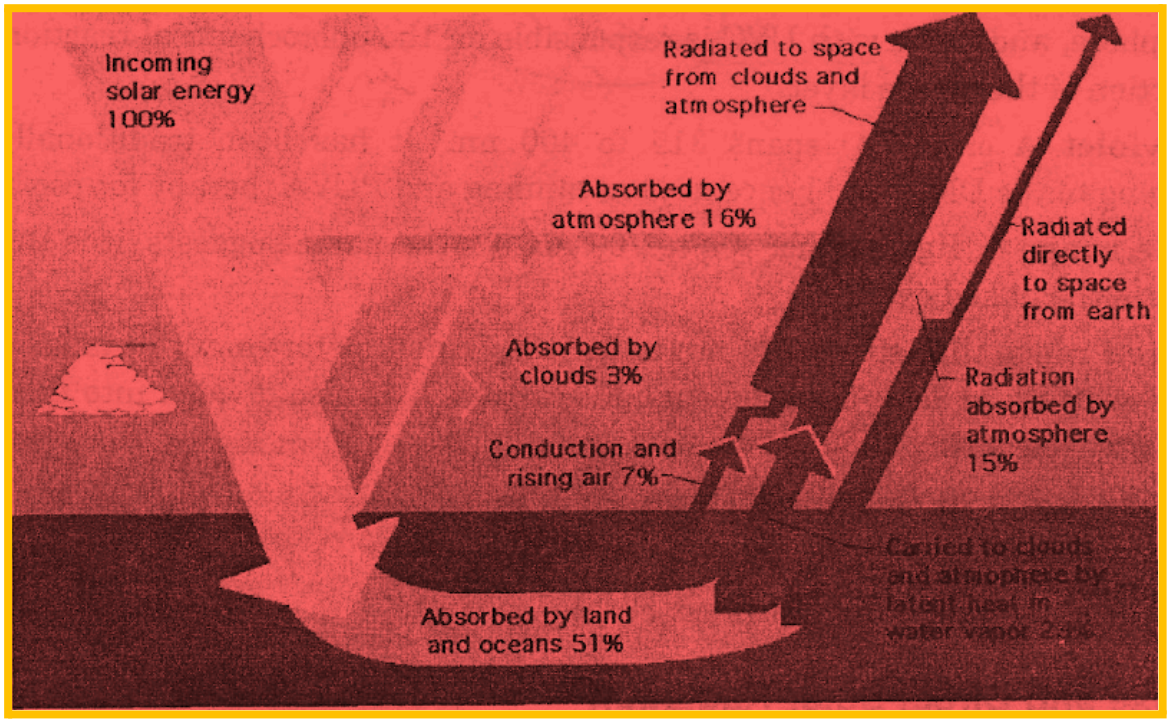
Sunlight absorbed by the oceans aids the land masses by keeping the surface at an average temperature of 14°C.

The conversion of solar energy into chemical energy via photosynthesis produces food, wood and the biomass from which fossil fuels are derived.

Solar radiation along with secondary solar resources such as wind and wave power, hydroelectricity and biomass account for 99.97% of the available renewable energy on Earth. The total solar energy absorbed by Earth's atmosphere, oceans and land masses is approximately 3,850 zettajoules (ZJ) per year.

The amount of solar energy reaching the surface of the planet is so vast that in one year it is about twice as much as it will ever be obtained from all of the Earth's non-renewable resources like coal, oil, natural gas, and mined uranium combined.

4.Earth's Energy Budget:



5.Earth's Energy Absorption & Reflection:

The solar constant is the amount of incoming solar electromagnetic radiation per unit area, measured on the outer surface of Earth's atmosphere in a plane perpendicular to the rays.

The solar constant includes all types of solar radiation, not just the visible light.

Thus, for the whole Earth (which has a cross section of 127,400,000 km^2), the power is 1.740 x 10^{17} W, plus or minus 3.5%. The solar constant does not remain constant over long periods of time.

The spectrum of the Sun's solar radiation is close to that of a black body with a temperature of about 5,800 K.

About half part lies in the visible short-wave part of the electromagnetic spectrum and the other half mostly in the near-infrared part. Some also lies in the ultraviolet part of the spectrum.

When ultraviolet radiation is not absorbed by the atmosphere or other protective coating, it can cause a change in human skin pigmentation.

6.Composition of Solar Radiation:

The spectrum of electromagnetic radiation striking the Earth's atmosphere is 100 to 10^6 nanometres (nm). This can be divided into five regions in the increasing order of wavelengths:

1.Ultraviolet-C or (UVC) range, which spans over a range of 100 to 280 nm. The term ultraviolet refers to the fact that the radiation is at higher frequency than violet light (and hence also invisible to the human eye). Owing to absorption by the atmosphere, very less rays reaches the Earth's surface (Lithosphere). This spectrum of radiation has germicidal properties and is used in germicidal lamps.

2.Ultraviolet-B or (UVB) range spans over a range of 280 to 315 nm. It is also greatly absorbed by the atmosphere, and along with UVC is responsible for the photochemical reaction leading to the production of the ozone layer.

3.Ultraviolet-A or (UVA) spans over a range of 315 to 400 nm. It has been traditionally held as less damaging to the DNA, and hence is used in tanning and PUVA therapy for psoriasis.

4.Visible range of light spans from 400 to 700 nm. As the name suggests, it is this range that is visible to the naked eye.

5.Infrared range spans from 700 nm to 10^6 nm (1 millimetre). It is largely responsible for the warmth or heat that the sunlight carries. It is also divided into three types on the basis of wavelength:

1.Infrared-A: 700 nm to 1400 nm

2.Infrared-B: 1400 nm to 3000 nm

3.Infrared-C: 3000 nm to 1 mm

7.Diffuse and Direct Solar Radiation:

As sunlight passes through the atmosphere, some of it is absorbed, scattered, and reflected by the following:
1.Air molecules

2.Water vapour

3.Clouds

4.Dust

5.Pollutants

6.Forest fires

7.Volcanoes

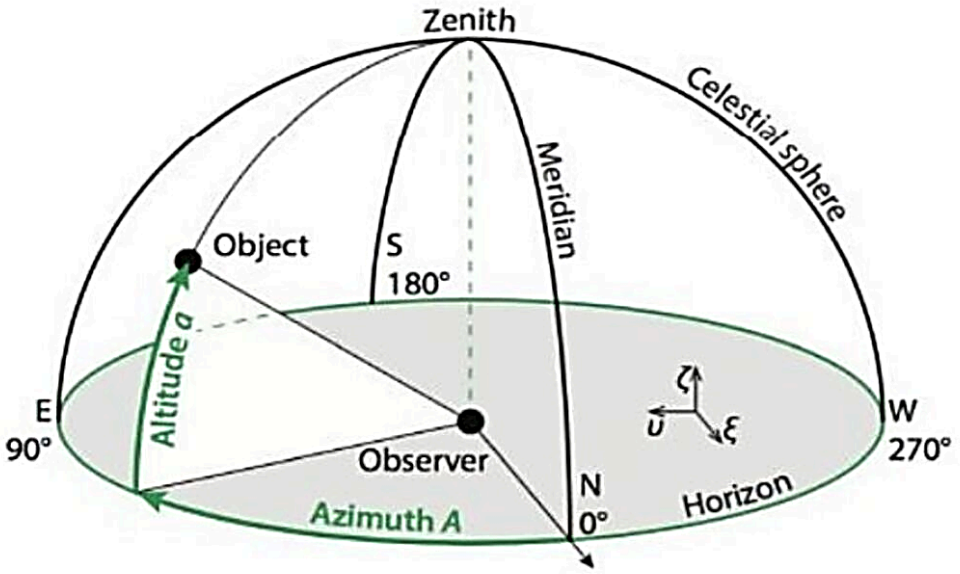
This is called diffuse solar radiation. The solar radiation that reaches the Earth's surface without being diffused is called direct beam solar radiation. The sum of the diffuse and direct solar radiation is called global solar radiation. Atmospheric conditions can reduce direct beam radiation by 10% on clear, dry days and by 100% during thick, cloudy days.

8.Solar Radiation Geometry:

Solar energy is one of the most prominent renewable energy source in India. The solar panel's output efficiency at max is 22% and it heavily depends upon various internal and external factors. To begin with, solar angles are one of the crucial external factors to be considered while installing the photo voltaic panel on any roof. Hence, our main focus will be on these different types of solar angles.

The 7 basic solar angles are:

1.Azimuth Angle:



As we all know heaven's objects such as the Sun, the Moon and the stars are far away from the Earth. Therefore, to simplify the motion measurements of these objects, the imaginary sphere is used, which contains an arbitrary radius and it is concentric towards the Earth. Coming back to the azimuth angle, it is calculated by measuring the angle between North (0 degree) and the altitude angle. Besides, the altitude angle (a) is nothing but the angular elevation of the object above the observer's horizon. The range of angular elevation varies from (-90 to 90 degrees), where -90 stands for the object being below the horizon and thus not visible. In this case, the observer's horizon can also be called the fundamental plane. On the other hand, the azimuth angle can be calculated in 4 main constituents, such as A= 0, 90,

180, 270 degrees which correspond respectively to North, East, South, and West.

Azimuth angle can be calculated as:

$$\cos(A) = \frac{[Sin(a) * \sin(L)] - \sin(D)}{\cos(a) * \cos(L)}$$

Where,

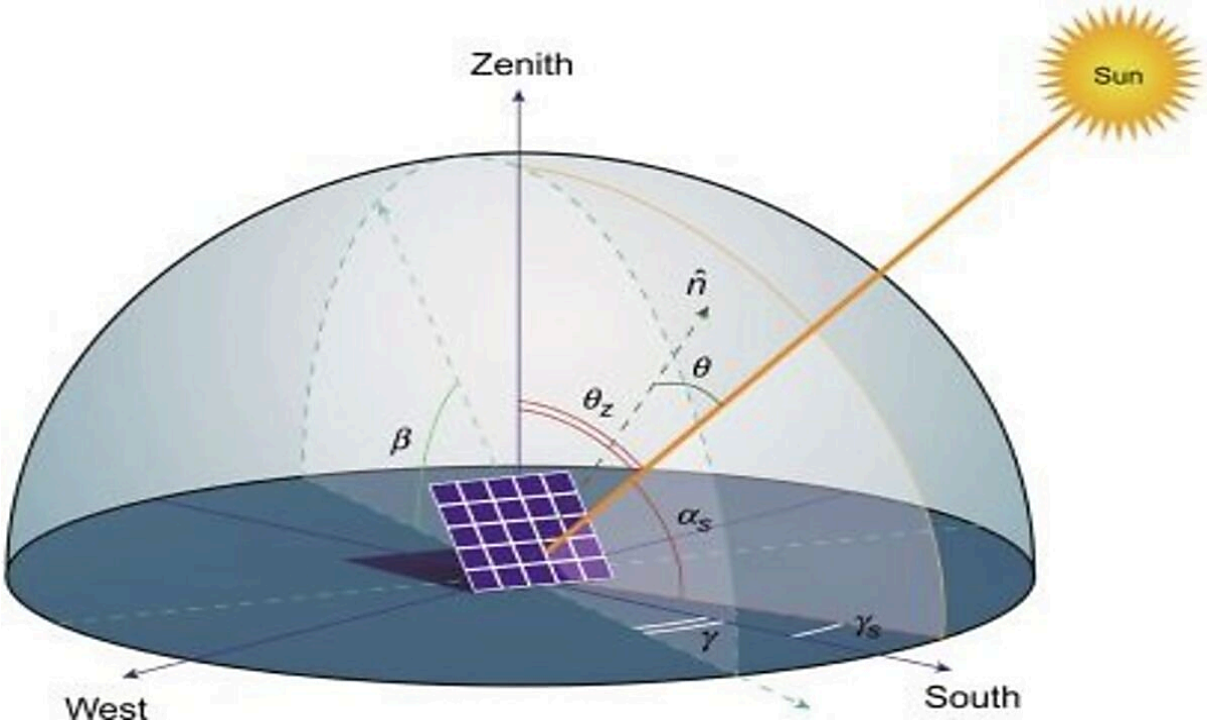
A = azimuth angle

a = altitude angle

L = latitude

D = declination angle

2.Zenith Angle:



Zenith in astronomy terms means the point in sky which is exactly above you, no matter wherever you stand on the Earth. Keeping that in mind, the zenith angle becomes an angle between the zenith and the Sun. It works complementary with the elevation angle, in which angular motion is calculated between the horizon and the Sun. As mentioned earlier these two angles work in a complementary manner, hence cosine of the one equals the sine of the other.

Zenith angle can be calculated as:

$$\cos(\theta_z) = \sin(D)\sin(L) + \cos(\mathrm{D})\cos(L)\cos(\omega)$$

Where,

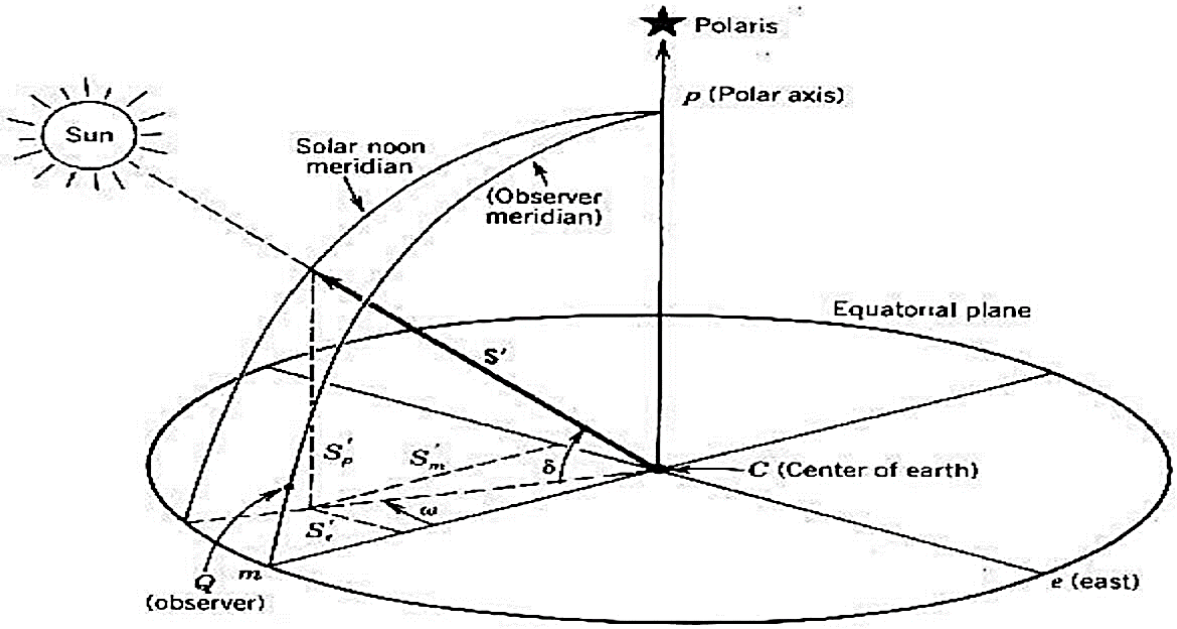
θ_z = zenith angle

D = declination angle (It varies from 23°27′ on June 21st to −23°27′ on December 22nd)

L = latitude (at noon $\cos(\omega) = 0\ and\ \theta z = L - D$

ω = hour angle (at sunrise or sunset [except North and South pole] $\cos(\theta z) = 0\ and\ \omega = \frac{N_d}{2}$)

3.Hour Angle:



The hour angle is the angular distance between the line of longitude (also known as the meridian) of the observer and the line of longitude which contains the Sun. Besides, the hour angle becomes zero at noon. Also, whenever the observers longitudinal plane contains the Sun, the hour angle increases by 15 for every hour.

The hour angle can be calculated as:

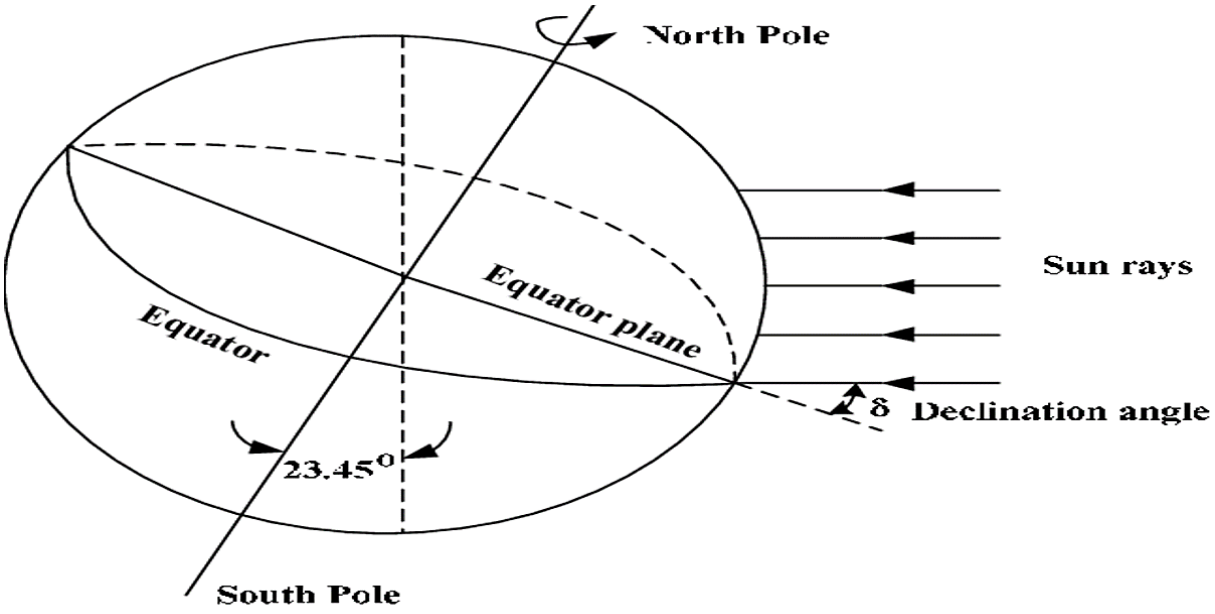
$$\omega = (ST - 12) * 15$$

(It varies from negative before the solar noon to positive after the solar noon)

For Example:

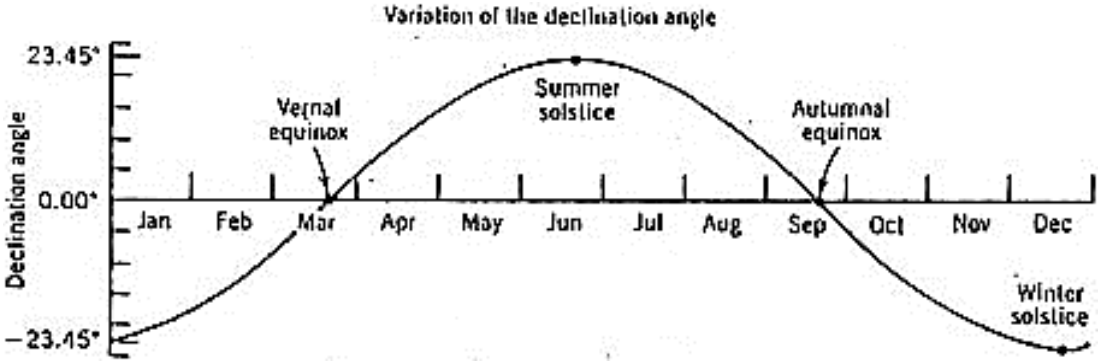
Time	10:00am	11:00am	12:00pm	1:00pm	2:00pm
Hour angle	$\omega = -30°$	$\omega = -15°$	$\omega = 0°$	$\omega = 15°$	$\omega = 30°$

4.Declination Angle:



It is the angle made between the ray of the Sun, extended to the centre of the Earth, and the equatorial plane of the Earth. Whenever the rays of Sun reach to the centre of the Earth via northern hemisphere, the declination angle becomes positive and whenever it reaches from the southern hemisphere, it becomes negative.

In the summer solstice, the value of solar declination angle is marked as +23.5 degrees and during the winter solstice, it is marked as -23.5 degrees. Hence the solar declination angle varies from -23.5 < d <23.5 degrees.



The declination angle can be calculated as:

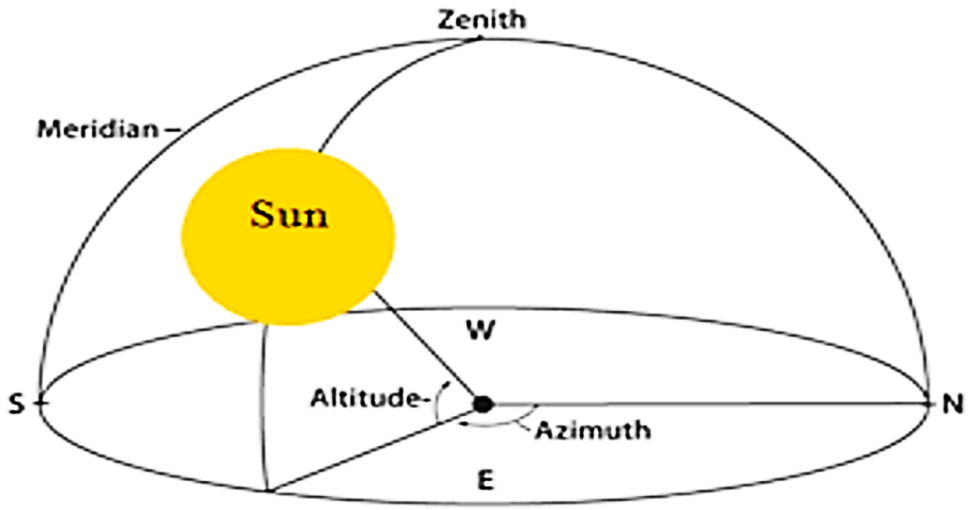
$$sin\ \delta = 0.39795.\cos\left[0.98563 * (N - 173)\right]$$

Where,

δ = declination angle

N = number of days since January 1st.

5.Altitude Angle:



The solar altitude angle is the angular distance between the rays of Sun and the horizon of the Earth. The solar altitude angle varies because of the three main factors which are, the time of the day, the time of the year, and the latitude of the Earth respectively.

The altitude angle can be calculated as:

$$\sin a = [\cos(L) * \cos(D)] * \cos(\omega)] + [\sin(L) * \sin(D)]$$

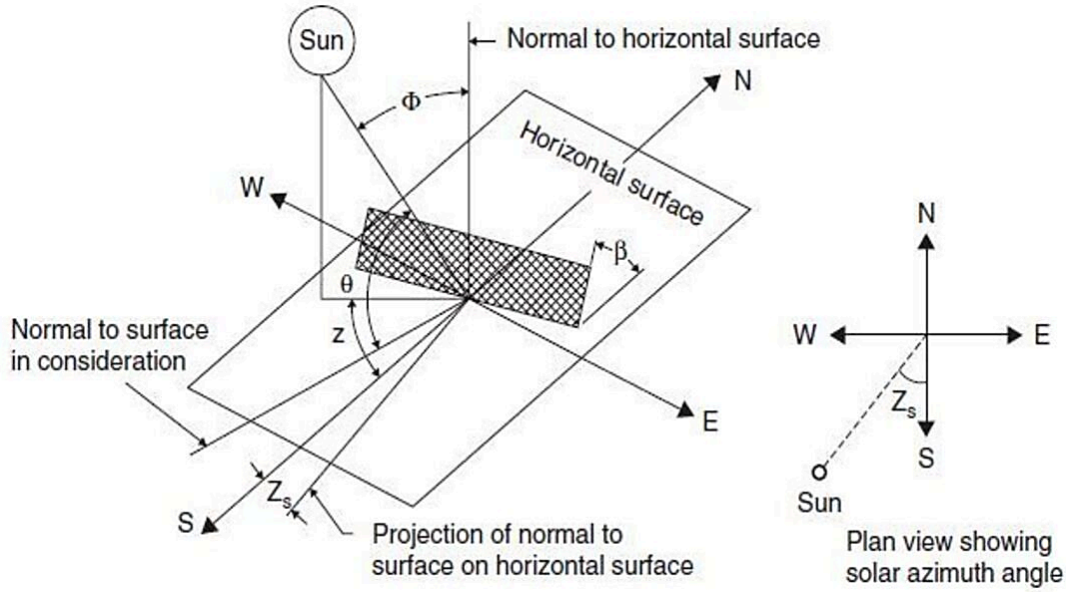
Where,

D = declination angle (It varies from $23°27'$ on June 21st to $-23°27'$ on December 22nd)

L = latitude (at noon $\cos(\omega) = 0\ and\ \theta z = L - D$

ω = hour angle (at sunrise or sunset [except North and South pole] $\cos(\theta z) = 0\ and\ \omega = \frac{Nd}{2}$)

6.Incidence Angle:



With respect to solar energy systems: The angle of the sun's ray renders a line perpendicular to the earth. For example, the Earth directly facing the Sun has an angle of zero incidence, and the surface parallel to the Sun (such as sunrise reaching a horizontal rooftop) has an angle of 90 degree. Sunlight at 90 degree angle of incident tends to be absorbed, while lower angles tend to be reflected.

Incidence angle can be calculated as:

$$\cos\theta = \sin L\, sin\delta\, cos\beta - cosL\, sin\delta\, sin\beta\, cosZ_s + cosL\, cos\delta\, cos\omega\, cos\beta + sinL\, cos\delta\, cos\omega\, sin\beta\, cosZ_s + cos\delta\, sin\omega\, sin\beta\, sinZ_s$$

Where,

β = surface tilt angle from the horizontal

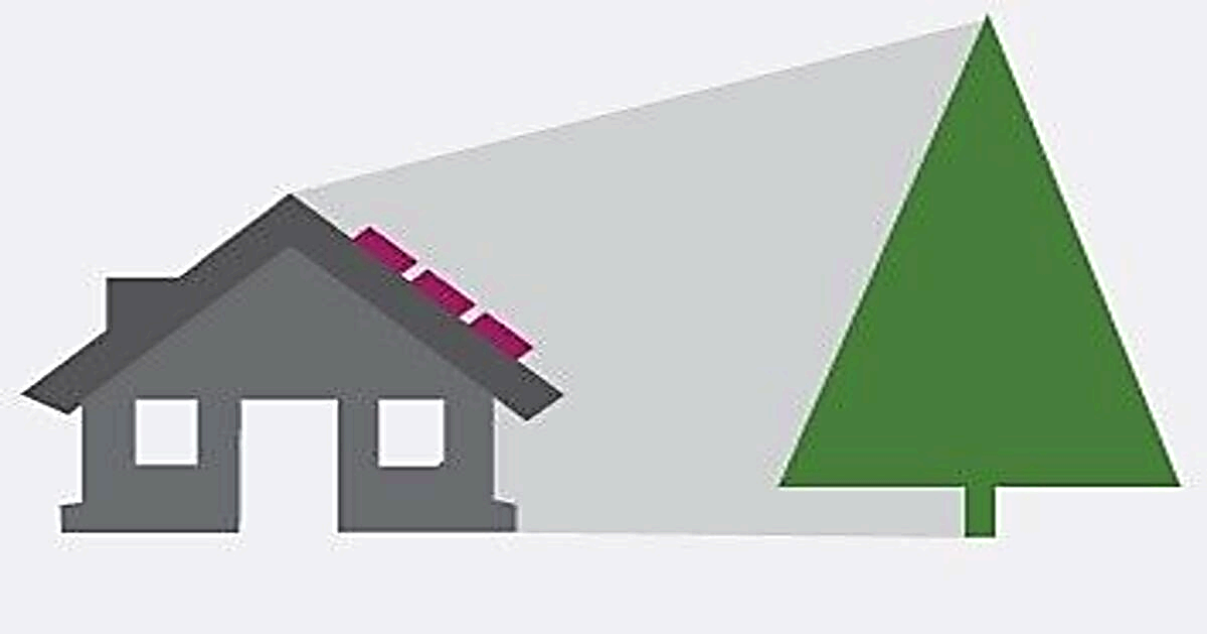
Z_s = surface azimuth angle, the angle between the normal to the surface from true south, westward is designated as positive

δ = declination angle

L = latitude

ω = hour angle

7.Shadowing:



While installing the solar panels on the roof or the facade, the shadowing factor plays a crucial role. It is the shadow projected by the surrounding objects on the panel. Besides, it is necessary to be aware of the shadow cast, as it changes daily.

The shadowing factor can be regulated by a “solar profile angle” and it can be calculated as:

$$\tan(p) = \frac{\tan(a)}{\cos(\theta_z - \theta_{zs})}$$

Where,

p = solar profile angle

a = altitude angle

θ_z = azimuth angle

θ_{zs} = surface azimuth angle

Conclusion:

In conclusion, to enhance this investigation and decrease possible errors, a number of angles must have been done. There might be a stand that is set at the right angle. That would make the angle accurate and not a few degrees off, which could interrupt the collection of energy. There could also only be experiments on clear days with almost no clouds to prevent clouds from changing the results to enhance the experiment. There could be multiple solar panels set at different angles to further analyse what angle of a solar panel would work best. The solar panels would then be left out all day and then see which angle, after 24 hours, gathered the greatest amount of energy.

9.Principle of Solar Energy Conversion:

The basic principle for the heat conversion is greenhouse effect. This effect is used to maintain temperature for the growth of some plants in cold climates. Most of the energy coming from sun is in form of light. It contains shortwave radiation which is not visible to eye. When sunrays fall on earth surface, it is heated due to absorbed radiation. The heated surface emits heat back. However, some of the gases in the lower atmosphere acting like glass in greenhouse allow solar radiation to fall on earth but does not allow the earth to reradiate the heat into space. The net result is the heating of Earth’s surface and this phenomenon is known as "Greenhouse Effect".

The same principle is used to generate heat from the solar radiation. It is carried out by using collectors. Usually, flat plate collectors are used for it. It is black painted metal surface covered with ordinary glass. When temperature of plate increases it radiates back. But it is trapped due to glass surface and reradiates on the plate. This phenomenon goes on until equilibrium temperature is reached. At equilibrium temperature, energy gain by absorption of the visible light is exactly balanced by the loss of energy through infrared emission of the glass plate.

CONCEPT:

Our project focuses on complete independence of an entire complex from fossil fuels. Our project takes into consideration the fact that there are very limited reserves of fossil fuels. The increasing cost also poses a huge stress on our economy. The concept is to utilize non-conventional energy sources of sun and bio-fuel for the entire energy needs of the complex. We are going to use solar energy for electricity production, space and water heating. Biogas production in a centralised biogas production unit is applied. A large capacity, low pressure pumping unit of biogas will be used to supply biogas at a sufficient rate to every household in the complex, replacing L.P.G. The biogas can also be used to generate electrical energy by running a standby generator using an I.C. engine, if the power produced by the solar power generation system is insufficient. The use of solar water heater saves a lot of energy which otherwise would have been wasted in running electric geysers. The solar water heaters will also be centralised. Other energy conservation techniques will be incorporated into the entire complex. Another advantage of this system will be that the costs incurred by the people in this system will only be of maintenance. The use of power saving tools like CFLs, LCD screens and other energy conservation tools must be kept in mind. The concept idolizes completely economical energy.

The Renewable Energy Resource Base (Exajoules Per Year):

	Theoretical	Current Use (2001)	Technical Potential
Hydrological Energy	9	50	147
Biomass Energy	50	>276	2900
Wind Energy	0.12	640	6000
Solar Energy	0.1	>1575	3900000
Geothermal Energy	0.6	not estimated	not estimated
Ocean Energy	not estimated	not estimated	7400
Total Energy	60	>1800	>4000000

The above table shows the capability of renewable energy in future. Thus, we can understand the need of using this energy in place of conventional energy sources.

The gradual replacement of energy sources is necessary as we cannot completely stop them.

There is no shortage of solar-derived energy on Earth. Indeed, the storages and flows of energy on the planet are very large relative to human needs.

Annual photosynthesis by the vegetation in the United States is 50 billion GJ, equivalent to nearly 60% of the nation's annual fossil fuel use.

The amount of solar energy intercepted by the Earth every minute is greater than the amount of energy the world uses in fossil fuels each year.

The energy in the winds that blow across the United States each year could produce more than 16 billion GJ of electricity, more than one and one-half times the electricity consumed in the United States in 2000.

Tropical oceans absorb 560 trillion gigajoules (GJ) of solar energy each year, equivalent to 1,600 times the world's annual energy use.

PROCEDURE OF PROJECT:

Installation of Solar Panels:

Solar cells are often electrically connected and encapsulated as a module. We use these modules in a little more than the required number so that the charging of the batteries will be sufficient. Solar cells are also usually connected in series in modules, creating an additive voltage. Connecting cells in parallel will yield a higher current. Modules are then interconnected, in series or parallel, or both, to create an array with the desired peak DC voltage and current. The required peak voltage and current is predetermined using calculations and estimations. The alignment and arrangement of panels must be such that there will be enough incident light on their surface for them to work satisfactorily. The directioning of panels must be correct and their surface must be horizontal or convex with east to west facing.

The power output of a solar array is measured in watts or kilowatts. In order to calculate the typical energy needs of the application, a measurement in watt-hours, kilowatt-hours or kilowatt-hours per day is often used. A common rule of thumb is that average power is equal to 20% of peak power, so that each peak kilowatt of solar array output power corresponds to energy production of 4.8 kWh per day (24 hours x l kW x 20% = 4.8 kWh).

To make practical use of the solar-generated energy, the electricity is most often fed into the electricity grid using inverters (grid-connected photo voltaic systems), in standalone systems, batteries are used to store the energy that is not needed immediately.

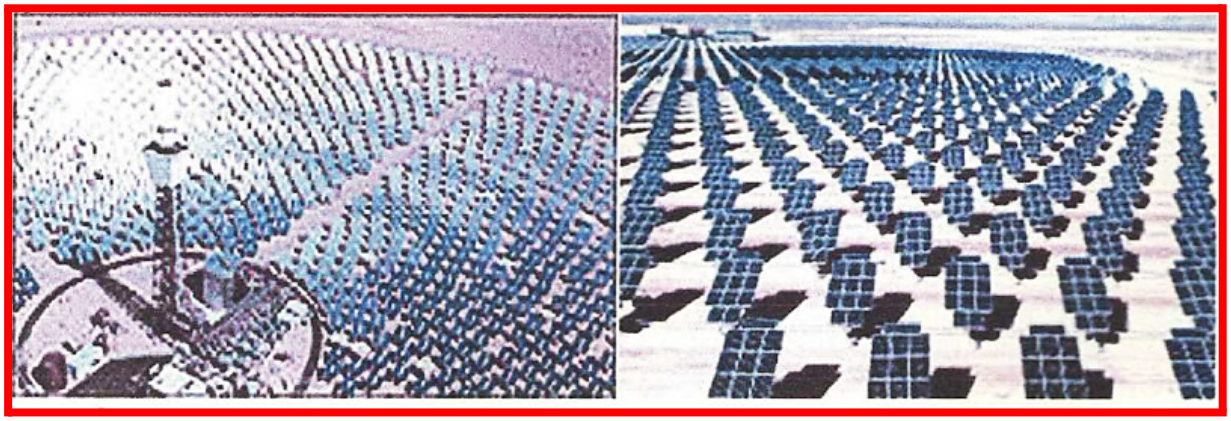
Solar energy is not available at night, and energy storage is an important issue because modern energy systems usually assume continuous availability of energy.

Thermal mass systems can store solar energy in the form of heat at domestically useful temperatures for daily or seasonal durations. Thermal storage systems generally use readily available materials with high specific heat capacities such as water, earth and stone. Well-designed systems can lower peak demand, shift time-of-use to off-peak hours and reduce overall heating and cooling requirements.

Phase change materials such as paraffin wax and Glauber's salt are another thermal storage media. These materials are inexpensive, readily available, and can deliver domestically useful temperatures (approximately 64 °C). The "Dover House" (in Dover, Massachusetts) was the first to use a Glauber's salt heating system, in 1948.

Solar energy can be stored at high temperatures using molten salts. Salts are an effective storage medium because they are low-cost, have a high specific heat capacity and can deliver heat at temperatures compatible with conventional power systems. The Solar Two used this method of energy storage, allowing it to store 1.44 TJ in its 68 m^3 storage tank with an annual storage efficiency of about 99%.

Off-grid photo voltaic systems have traditionally used rechargeable batteries to store excess electricity. With grid-tied systems, excess electricity can be sent to the transmission grid. Net metering programs give these systems a credit for the electricity they deliver to the grid. This credit offsets electricity provided from the grid when the system cannot meet demand, effectively using the grid as a storage mechanism.



WORKING MODEL DATA:

The solar panel is installed and exposed to the solar radiation to produce direct current which is then fed to the electric motor. The function of electric motor is to receive the direct current from solar panel and then convert that current into the mechanical movement.

This mechanical movement is in rpm. The electric motor used in our working model is of 200 rpm. This rpm is not suitable for the lifting purpose. And also, the torque accomplished with that rpm is also very low. For lifting application, the torque should be high enough.

To achieve the above requirements, we used gearbox. The gearbox contains 6 number of gears. These gears are helical type of gears and then it is assembled using compound type of gear box. The 200 rpm acts as an input rpm and it is finally converted into 60 rpm.

For Compound Gears:

(N2/N1) x (N4/N3) x (N6/N5) = (T1/T2) x (T3/T4) x (T5/T6)

Where, N1, N2, N3, N4, N5 and N6 are rpm of six gears respectively and T1, T2, T3, T4, T5 and T6 are number of teeth of the 6 gears respectively.

Now, the velocity ratio can be given by:

Velocity Ratio (VR) = Noutput/Ninput

Now, we have velocity ratio (T1/T2):

T1=9

T2=30

VR=9/30

Therefore, 9/30=(Noutput /200)

Implies, Noutput=60

Therefore, the torque measured was 5kg/cm^2

This reduced rpm is successfully utilized for load lifting purpose. Another desirable advantage of the gearbox is that it made the solar load lifter of self-braking capacity. It means that we can hold the load at any desired position. This self-braking occurs when the static friction angle is greater than the helical angle and due to this back driving is also prevented.

The rope used in this working model is of cotton. This rope is fabricated by using synthetic bond and then it is subjected to destructive testing for tensile application.

This wire rope is then allowed to pass through the section rollers. Total 3 section roller are used. The specifications are standard one and is ¾ section roller.

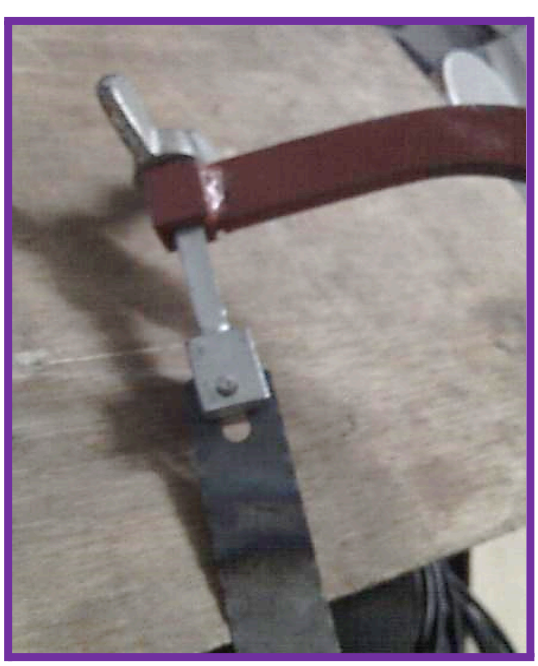
These section rollers ensure the frictionless and jerk free movement of rope.

MANUFACTURING PROCESS:

1.Manufacturing of Components:

The manufacturing parts of the working model is as given follows:

1.Base foundation: Base foundation is made up of wooden platform. The main advantage of wooden platform is that the wooden piece is easy to machine and it is enough to support the load lifting structure. Initially it is cut in the desired shape and then the drills are produced according to structure. The cutting of the plywood is done by the professional carpenters in order to obtain the desired accuracy.





2.Structure: While choosing the suitable material for the structure, there are two conditions that have to be satisfied to ensure the firm support of load lifter. The aluminium was chosen finally which was satisfying the two conditions. Square rods of aluminium are first cut according to the dimensions and then the holes were produced for fixations.

3.Gears: Lifting the load with the help of motor was our aim. To do that it, was essential that the motor should have high starting torque. To obtain the high torque there are two options with us. Use high torque motor or use gearbox. The available power was limited and high torque electrical motors required high initial current. To overcome this problem, we decide to use gearbox. Instead of using high torque motor we use high rpm motor and the high rpm is then converted into the low rpm so that we can achieve the high starting torque. Motor shaft is further attached to gearbox.

2.Purchased items:

1.Brackets: The brackets are used as a bonding member in between the wooden platform (horizontal and vertical).

2.Electric motor: Electric motor is operated on direct current. This motor is chosen according to the solar panels specification so that it is possible to utilize the full power. The high rpm of motor is transferred into the low rpm and hence high torque can be achieved. This high torque with low rpm is further utilized to lift the load. The motor is used to convert the collected direct current into the mechanical movement.



3.Switch: The switch is required to trigger the lift in upward and downward direction. To carry out the determined operation, the DP/DT type of switch is most suitable. This type of switches are reversible in nature.

4.Cable: The cable is nothing but rope that lifts the load. The cotton rope with polyester is found to be satisfactory and safe. Still to increase the factor of safety, four layers of ropes are taken and then it is bound together with the help of synthetic resin. The disadvantage of this resin is that these bonds are not water proof.

5.Rollars: To ensure the friction free movement of thread with structure, a number of rollers are provided. In other words, the rollers are used there for efficient working of thread.

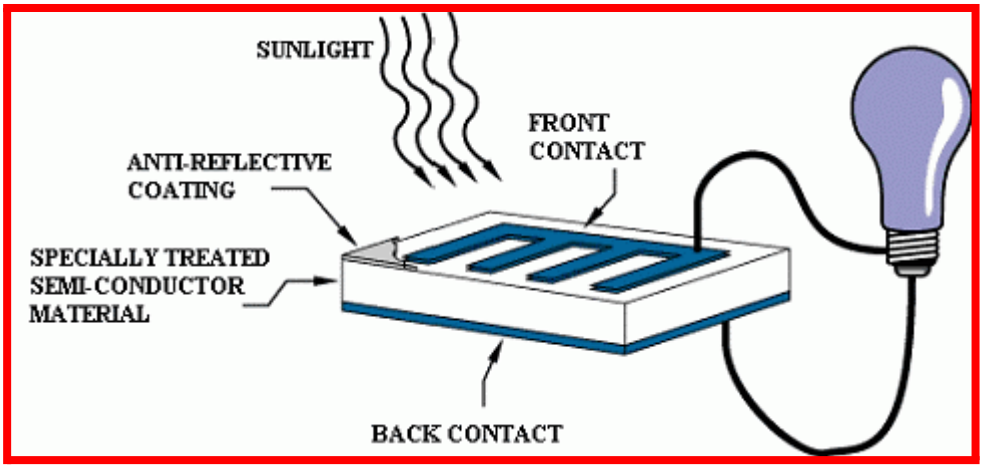
6.Hook: Hook is the member that is tied with the rope at one side and on the other end, it is kept open to attach the loads. The spring-operated enclosure is given to avoid the escapement of the load while raising and lowering.

7.Photovoltaic Cell:

Photovoltaics is the direct conversion of light into electricity at the atomic level. Some materials exhibit a property known as the photoelectric effect that causes them to absorb photons of light and release electrons. When these free electrons are captured, an electric current result that can be used as electricity.

The photoelectric effect was first noted by a French physicist, Edmund Becquerel, in 1839, who found that certain materials would produce small amounts of electric current when exposed to light. In 1905, Albert Einstein described the nature of light and the photoelectric effect on which photovoltaic technology is based, for which he later won a Nobel prize in physics. The first photovoltaic module was built by Bell Laboratories in 1954.

It was billed as a solar battery and was mostly just a curiosity as it was too expensive to gain widespread use. In the 1960s, the space industry began to make the first serious use of the technology to provide power aboard spacecraft. Through the space programs, the technology advanced, its reliability was established, and the cost began to decline. During the energy crisis in the 1970s, photovoltaic technology gained recognition as a source of power for non-space applications.



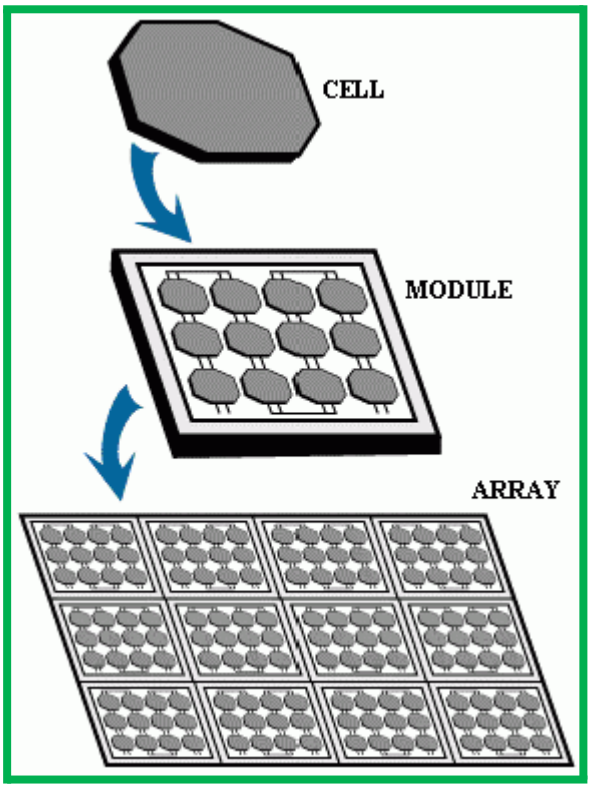
The diagram above illustrates the operation of a basic photovoltaic cell, also called a solar cell. Solar cells are made of the same kinds of semiconductor materials, such as silicon, used in the microelectronics industry. For solar cells, a thin semiconductor wafer is specially treated to form an electric field, positive on one side and negative on the other.

When light energy strikes the solar cell, electrons are knocked loose from the atoms in the semiconductor material. If electrical conductors are attached to the positive and negative sides, forming an electrical circuit, the electrons can be captured in the form of an electric current, i.e., electricity. This electricity can then be used to power a load, such as a light or a tool.

A number of solar cells electrically connected to each other and mounted in a support structure or frame is called a photovoltaic module. Modules are designed to supply electricity at a certain voltage, such as a common 12 volts system. The current produced is directly dependent on how much light strikes the module.

Multiple modules can be wired together to form an array. In general, the larger the area of a module or array, the more electricity that will be produced. Photovoltaic modules and arrays produce direct-current (dc) electricity. They can be connected in both series and parallel electrical arrangements to produce any required voltage and current combination.

The output of these cells is connected to the electric motor.



3.Technical Specification of the Solar Photovoltaic Cell Used:

MODEL: SLP05-12

RATED POWER: 5 W

RATED VOLTAGE: 17 W

RATED CURRENT: 0.45 A

OPEN CIRCUIT VOLTAGE: 21.6 V

SHORT CIRCUIT CURRENT: 0.34 A

STANDARD TEST CONDITION: 1000 W/m^2, 1.5 A-m & 25 C

4.Manufacturing Product:

Name of the Part: Plywood

Quantity: 1

Material: Wood

Overall Sizes:

Length=700 mm

Diameter=160 mm

5.Total Time Consumption:

Serial Number	Operation	Feed	Tool	Time in Minutes
1.	Cutting the wood according to the designed shape	Manual	Hack saw blade	50
2.	Drill 6 component having 3.5 mm diameter	Manual	Drill, Boring	30
3.	Applying the coating on the plywood	Manual	----------	20
4.	File of length of 60 mm rod. Mark of the centre.	Manual	File, Punch	20 minutes each
TOTAL TIME IN MINUTES				140

6.Cost Estimation:

Serial Number	Name of Equipment	Quantity	Cost (In Rupees)
1.	Hook	1 Nos.	10
2.	Solar Panel	1 Nos.	4800
3.	Plywood	1 Nos.	50
4.	Electronic Motors	1 Nos.	420
5.	Nut & Bolt	8 Nos.	30
6.	Circuit Panel	1 Nos.	10
7.	Wire	1m	50
8.	Clips	1 Nos.	20
9.	DPDT Switch	2 Nos.	40
10.	Bracket	1 Nos.	20
11.	Gears	8 Nos.	60
12.	Aluminium Bars	1.5 m	180
13.	Roller Bearings	4 Nos.	80

7.Cost Estimation Data:

The cost estimation of the project is based on the following data:

Raw Material Cost:

Weight of raw stock in kg multiplied by the market rate per kg.

Labour Charges for Machining:

Machining time in hours multiplied by the rate of machining per hour.

Summary of Cost:

Raw Material Cost: Rs. 5770

Labour Charges for Machining: Rs. 2112

Other Charges: Rs. 234

Transport Charges: Rs. 1580

Total Cost: Rs. 9696

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